

Complex Trigonometric and Related Functions

MA4410

Power Series Definitions (5.4.1)

Definition

For $z \in \mathbb{C}$, define

$$\sin z := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}, \quad \cos z := \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}.$$

Both series have infinite radius of convergence and define entire functions.

Definitions via the Complex Exponential (Consequences)

Definition

For $z \in \mathbb{C}$, define

$$\sin z := \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z := \frac{e^{iz} + e^{-iz}}{2}.$$

These definitions extend the real sine and cosine via the complex exponential.

Basic Properties

For all $z \in \mathbb{C}$:

- $\sin z$ and $\cos z$ are entire functions.
- $\sin(-z) = -\sin z$, $\cos(-z) = \cos z$.
- Euler formulas lead directly to identities.

Example: Evaluating $\sin(ix)$

Let $x \in \mathbb{R}$. Then

$$\sin(ix) = \frac{e^{-x} - e^x}{2i} = i \frac{e^x - e^{-x}}{2} = i \sinh x.$$

Complex sine naturally introduces hyperbolic functions.

Example

Solve $\sin z = 0$.

$$\sin z = 0 \iff e^{iz} = e^{-iz} \iff e^{2iz} = 1.$$

Thus

$$2iz = 2\pi ik \implies z = \pi k, \quad k \in \mathbb{Z}.$$

Same zeros as in the real case.

Definition

For $z \in \mathbb{C}$, define

$$\sinh z := \frac{e^z - e^{-z}}{2}, \quad \cosh z := \frac{e^z + e^{-z}}{2}.$$

These mirror the trigonometric definitions with z replacing iz .

For all $z \in \mathbb{C}$:

- $\cosh^2 z - \sinh^2 z = 1.$
- $\sinh(iz) = i \sin z.$
- $\cosh(iz) = \cos z.$

Example: $\cos(x + iy)$

Let $x, y \in \mathbb{R}$. Using exponentials,

$$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y.$$

Real and imaginary parts mix trigonometric and hyperbolic behavior.

- $|\sin(x + iy)|$ and $|\cos(x + iy)|$ grow exponentially as $|y| \rightarrow \infty$.
- Very different from bounded real sine and cosine.
- Important for understanding zeros and contour integrals.

Summary

- Complex trigonometric functions are defined via exponentials.
- Hyperbolic functions arise naturally from imaginary arguments.
- Many familiar identities persist, but geometric behavior changes.

Deriving the Formula for $\arcsin z$

Solve $\sin w = z$ using exponentials:

$$\frac{e^{iw} - e^{-iw}}{2i} = z.$$

Multiply by $2i$ and set $u = e^{iw}$:

$$u - \frac{1}{u} = 2iz.$$

Multiply by u :

$$u^2 - 2izu - 1 = 0.$$

Quadratic formula gives

$$u = iz \pm \sqrt{1 - z^2}.$$

Thus

$$w = -i \operatorname{Log} (iz \pm \sqrt{1 - z^2}).$$

Where Do the Cuts Come From? (Square Root)

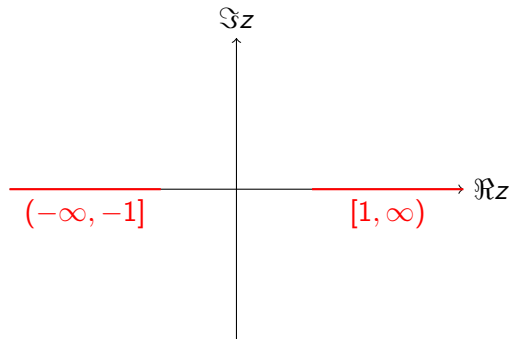
The term $\sqrt{1 - z^2}$ determines the branch geometry.

$$1 - z^2 = (1 - z)(1 + z).$$

Branch points occur where $1 - z^2 = 0$, i.e. at $z = \pm 1$.

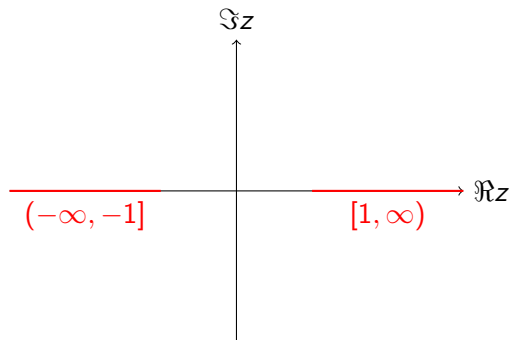
- Any branch of $\sqrt{w}$ requires a cut from $w = 0$ to ∞ .
- Under the map $w = 1 - z^2$, the real intervals $(-\infty, -1]$ and $[1, \infty)$ map to $[0, \infty)$.
- Hence those intervals must be removed to define a single-valued branch.

Branch Cuts for $\arcsin z$



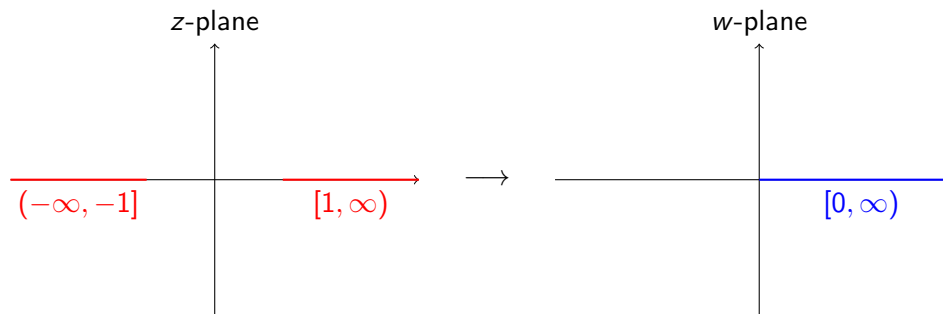
Branch cuts arise from $\sqrt{1 - z^2}$ and the logarithm.

Branch Cuts for "arctanh"



From $\text{Log} \left(\frac{1+z}{1-z} \right)$; singularities at $z = \pm 1$.

Mapping Picture: $w = 1 - z^2$



The intervals $(-\infty, -1]$ and $[1, \infty)$ map onto $[0, \infty)$, the standard branch cut for $\sqrt{w}$.